

Hernán Díaz Rodríguez
Department of Computing, University of Oviedo
Gijón, Spain
diazhernan@uniovi.es

5 October 2026

To the Editors,
Computers & Industrial Engineering

I submit for your consideration the manuscript *Genetic Programming Hyper-Heuristics for the Job Shop Scheduling Problem with Interval Durations: Robustness-Aware Interpretable Rules that Generalize Across Instance Sizes*, as a research paper.

In many manufacturing settings the duration of an operation is not known in advance, and a planner can only give a lower and an upper bound. The interval job shop scheduling problem models this situation, and its best solutions so far come from metaheuristics that search anew for every instance. The manuscript brings automated rule design to this problem. A genetic programming hyper-heuristic evolves dispatching rules that read both the worst-case value and the width of each uncertain quantity. The rules build a schedule in milliseconds and are short enough to be read and checked by a planner.

The methodological contribution has three parts. First, rules evolved on four instances improve on every constructive baseline on all seventy instances of the benchmark, and transfer without retraining to all size classes and to a second benchmark. Second, ablations show where interval information pays off: not in the expected makespan, but in the predictability of the schedule. An objective that penalizes the width of the predicted makespan yields schedules whose execution departs less from the prediction, with a single weight that sets the trade-off. Third, a comparison at matched budget with a genetic algorithm, reimplemented on the same decoder, shows up to which computational budget the evolved rule is the better choice, and how that budget grows with instance size.

The work fits the journal because it answers questions that a production planner faces: which rule to deploy when schedules must be built or rebuilt quickly, how much expected makespan to give up for a more reliable schedule, and when a per-instance search is worth its cost.

I confirm that the manuscript is original, that it has not been published elsewhere, and that it is not under consideration by another journal. An earlier version of this study is available as a preprint (SSRN, doi:10.2139/ssrn.7214286).

The complete source code is openly available in Zenodo (doi:10.5281/zenodo.21716972; version 2.0, which accompanies this manuscript: doi:10.5281/zenodo.23156759): the hyper-heuristic, the baselines and the genetic algorithm, together with a test that re-derives the deposited results from scratch. The deposit also holds the instances, the evolved rules and the result files behind every table and figure. A statement on the use of generative AI in the preparation of the manuscript appears directly before the references, as the journal's policy requires. In keeping with the double-anonymous review process, the author's identity appears only in the title page and in this letter.

Thank you for considering this work.

Yours sincerely,

Hernán Díaz Rodríguez
Corresponding author